

APLO: A Modular Adaptive Pedagogical Learning Orchestration Ontology for Neurosymbolic Educational Systems

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Abstract

Education is poised for a transformative shift with the emergence of increasingly adaptive and personalized artificial intelligence systems. Yet, realizing this vision requires explicit representations of how pedagogy, learner context, assessment and feedback interact over time. Structured knowledge provides a promising foundation for grounding adaptive learning experiences in established pedagogical frameworks while responding to evolving learner goals, preferences, challenges and awareness. However, existing ontologies predominantly represent curriculum resources, competencies or recommendation targets, offering limited support for representing the processes through which instruction is organized, evaluated and adapted. Adaptive pedagogical systems additionally require explicit representations of instructional structure, learning objectives and situations, evaluation and feedback processes and temporally evolving learner awareness. In this work, we address this modeling gap through a modular ontology for pedagogical, learning objective, learner and evaluation modeling. We evaluate the ontology and conclude how modular representations of pedagogical orchestration can bridge curricula to learner-aware instructional processes, supporting future neurosymbolic educational architectures.

Keywords

Pedagogical Framework, Learning & Instruction Strategy, AI Tutor, Educational Ontology

1. Introduction

Generative AI (GenAI) has rapidly expanded the range of computational support available for teaching and learning processes. Recent work examined large language models (LLMs) across tutoring, assessment, feedback, content generation and personalized learning, while systematic reviews have also identified concerns related to reliability, over-reliance, fairness, privacy and quality of educational use [1, 2, 3, 4, 5]. As these developments extend beyond a single class of large models, educational systems increasingly draw on heterogeneous generative and language-model capabilities, including smaller deployable models and hybrid AI architectures, which create new opportunities to adapt explanations, resources, feedback and interaction to a learner and learning context. Yet generative capability alone does not determine how instruction should be organized or adapted, as a true adaptive system also needs to represent (a) what a learner aims to achieve, (b) learner’s current understanding, (c) which pedagogical approach applies, (d) what evidence an assessment produces, (e) what feedback follows and (f) how these conditions change over time.

We refer the coordination of all these concerns as *adaptive pedagogical orchestration* with context-sensitive organization and adjustment of pedagogical structures, techniques, objectives, resources, evaluations and feedback in response to activity, goals, characteristics and evolving awareness observations. This problem differs from resource generation or recommendation alone as it often requires the

WOP’26: 17th Workshop on Ontology Design and Patterns, October 25–29, 2026, Bari, Italy

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ability to relate instructional processes to learner context and assessment evidence over time. Educational knowledge graph (KG) efforts have shown the value of structured knowledge for personalized learning, curricula design, concept mapping and recommendations while curricula-focused ontologies further show how explicit semantic models can represent interlinks with learning materials [6, 7, 8]. However, representing these educational entities and relationships does not explicitly provide the processes through which instruction is structured, evaluated or adapted.

This distinction leaves a more specific representational gap as adaptive orchestration (pedagogically) requires additional semantics with ordered pedagogical structures, relations among objectives, instructional techniques, assessment results, feedback provenance and their changes over time. If these concerns remain distributed across isolated schemas or loosely connected data graphs, the explicit support for interpreting how an assessment result relates to an objective, how feedback arose from that result or how a learner observation contextualizes a subsequent instruction process becomes limited. Therefore, there is a compelling need for process-aware ontology models that represent these dependencies explicitly rather than treating pedagogy, learner, learning material and assessment data as independent silos.

These needs become more salient as AI-supported tutoring moves toward increasingly adaptive and conversational interaction. Yet the literature has also identified technical confounding and reliance concerns around such systems [3, 5, 9]. Neurosymbolic AI (NeSy) offers one direction by combining the capabilities of neural models with explicit symbolic knowledge and reasoning structures, enabling hybrid architectures in which pedagogical agents, LLMs and KGs combined can support adaptive, personalized learning [10]. In that context, Neurosymbolic AI-powered Pedagogical Agents (NaPAs) motivate a concrete representational requirement: *an agent cannot ground pedagogical adaptation only in curriculum but also account for instructional structure, learner tailoring, assessment evidence and feedback provenance*. The present work addresses this symbolic modeling problem not by defining a complete adaptive policy or implementing a NaPA but by developing an explicit ontology foundation to present any required information dependencies.

As such, we treat orchestration as a cross-module representation capability rather than a monolithic ontology class, reflecting the observation that pedagogical structure, learning objectives, evaluation processes and learner context form distinct modeling concerns while adaptive use depends on explicit interfaces among them. We therefore investigate the following research questions:

- RQ1.** What knowledge requirements are needed to represent adaptive pedagogical orchestration across instructional structure, evaluation and evolving learner context?
- RQ2.** How can reusable ontology design patterns be systematically adapted and composed into inter-operable modules that satisfy these requirements?
- RQ3.** To what extent does the ontology support its requirements when instantiated as a KG?

To address these questions, we developed a modular ontology and reusable ontology design patterns (ODPs) with three major contributions: (i) developing a modular ontology for adaptive pedagogical orchestration that represented pedagogical technique structures, learning objectives, learner evaluations, awareness and context, (ii) documenting a requirements-driven process for selecting, adapting and composing reusable ODPs across these modules, while connecting them to shared concepts and resources (iii) evaluating the ontology through structural quality analysis, materialization and competency question (CQ) validation.

Rest of the paper is organized as follows. Section 2 reviews background and prior works around our motivations, educational knowledge representation and instructional process modeling. In Section 3, we describe our ontology specifications, methodology used and design choices alongside ODPs utilized. Section 4 presents our approach to instantiating and evaluating the ontology, while Section 5 deep dives into ontological implications and tradeoffs. In Section 6, we briefly conclude with key takeaways and future directions.

2. Background & Related Work

2.1. Curriculum and Educational Knowledge Representation

Educational KGs became a common way to represent learning concepts, resources, curricula and recommendations. Prior works identified applications in personalized learning, curricula design, concept mapping and educational content recommendation [6, 8, 11, 12]. For our purposes, this literature establishes curricula or resource-representation layers on which adaptive orchestration builds.

Several systems made this curriculum-centered direction concrete. EduKG modeled K-12 educational knowledge and resources through a fine-grained ontology and large heterogeneous graph with emphasis on sustainable construction and maintenance of textbook-derived knowledge [11]. EducOnto and EduKG modeled higher-education curricula and student profiles for university curriculum recommendation [12]. Other course recommendation work used matching student preferences, course descriptions and program structures in order to support course selection or proposed ontology-based approaches for curriculum development, especially in higher education [13, 14]. These showed that ontologies can represent educational structures and support curriculum or course-level decision making.

Other works focused more on specificity. Teaching KG used semantic constraints and higher-level structure to support instructional resources for teaching KG education [8]. Domain-specific efforts also modeled narrower educational structures, such as a computer science topics ontology intended to support educators and curriculum work with extensions to other competencies and disciplines [15, 16]. Similarly, few have built evidence- and competency-linked KGs for academic pathway recommendation, planning and decisions by linking with curriculum structures [17, 18]. These works stood relevant as they value explicit curriculum, topic and profile models but they did not make the interaction among pedagogy, evaluation, feedback or evolving learner-state updates central in modeling.

The CurrKG ontology provided the closest curricular foundation for current work and represented curricula and learning materials as an ontology for densely interlinking educational materials across topics and organizational structures. It further evaluated the ontology through materialization and CQs [7]. We extend that line of work as a curriculum-facing foundation and further model the scope toward adaptive pedagogical orchestration, therefore addressing the complementary gap: curriculum and material links alone do not represent how an adaptive system selects pedagogical structures, interprets assessment evidence, produces feedback and updates learner awareness over time.

Broader surveys support this distinction that educational ontologies already cover many entities, such as learning objects, resources, content recommendation, disciplines and learner profiles [19, 20] but also motivate integrating those entities into process-aware models when the target is adaptivity rather than static representation or recommendation.

2.2. Instructional Process Modeling

Instructional process modeling is central to our goal because it represents teaching and learning as organized activities than only as content structures. Some efforts modeled instructional-design theories and used formal representations to check, generate and search learning designs, showing that these instructional methods and learning-design theories can be represented as formal knowledge. Others proposed ontology-based modeling frameworks for educational technology design by including domain patterns for also representing instructional design knowledge [21, 22, 23, 24]. These are especially relevant because they also treated educational, instructional and learning design as a modeling problem.

Learning objectives form another bridge between curricula and instructional process representations. Few ontologically mapped learning objectives, standards and gameplay goals, showing that objectives can be aligned across educational and interactive domains [25, 26]. This motivated modeling learning objectives as explicit interface entities among pedagogy, curriculum, learner situation and assessment but our work differs from these efforts in modeling target. Prior works represented instructional

theories, educational-technology knowledge, learning designs or objectives [21, 22, 25], whereas we instead model interactions among learner awareness context, evaluation techniques and feedback.

2.3. Learner and Personalization Modeling

Personalization to learner modeling provides another important foundation for adaptive orchestration. Adaptive educational systems have long used learner models to capture learner characteristics, manage uncertainty from multiple data sources and support instructional adaptation over time [27, 28, 29]. However, recent works also proposed profile modeling use, describing how they represent learner-centered information for true personalization or modeled educational plus career-oriented recommendation knowledge with learning resources, goals, paths, user profiles and skill information [30, 31, 32]. These show that semantic models can support learner-aware recommendation decisions.

Open learner models (OLMs) add an important transparency requirement where open, inspectable modeling makes learner model information available to learners or teachers, enabling to inspect, interpret and sometimes negotiate the system’s representation of learner knowledge or state [33, 34]. This line of work is relevant to our ontology because adaptive orchestration not only requires a hidden learner profile but also explicit learner-state representations that can be connected to evidence, feedback and explanation. Prior work, with frameworks (e.g., SMILI) further organizing OLM interfaces around learner data representation, access and user interaction, has also argued that OLMs provide lessons for interpretable AI in education because the model value depends on the interpretability of the underlying representation, not only on the fluency of the interface [35, 36].

Recent work with LLMs renewed this need to model learners more broadly, discussing the opportunities and challenges to address learner goals, context and broader support needs, even more for assessment conditions [37, 38]. This whole-learner perspective aligns with our position to model goals, preferences, challenges, awareness and learning situations as symbolic structures than leaving them implicit in natural-language interaction.

Taken together, while most prior work provided strong foundations for curricula, instructional design, learning-objective alignments, OLMs or personalized recommendations, these areas usually model separate parts of the adaptive education problem, where: **i.** resource ontologies model what is taught and its organization, **ii.** instructional-designs model pedagogical methods, learning designs and educational-technology design knowledge or **iii.** learner models represent goals, preferences, profiles or recommendation contexts. Our work bridges across these concerns and connects all strands by:

- modeling adaptive pedagogical orchestration as modular, pattern-based units;
- explicitly relating pedagogical structures, learning objectives, learner situations, evaluation techniques, feedback responses, source information and temporally evolving learner awareness; and
- representing learner-centric adaptive tutor process-aware details in modeling.

This focus also follows prior work on NeSy AI (i.e., NaPAs) with symbolic grounding for adaptive educational agents [10].

3. APLO Ontology

Adaptive Pedagogical Learning Orchestration (APLO) ontology modeled the information dependencies needed to represent adaptive orchestration while preserving pedagogical intricacies. Instead of modeling a complete tutoring policy, recommender algorithm or NaPA, it modeled the entities and relations needed to connect pedagogical structures, learning objectives, evaluation evidence, feedback, learner context and awareness observations (including progression and degradation), to scope to prior work’s representational gap (as identified in Section 2).

Table 1

Excerpt of CQ groups and key notions

CQ family	Representative CQ	Key notions
Pedagogical organization	What pedagogical element follows another pedagogical element?	PedagogicalFramework, PedagogicalElement
Objective alignment	What pedagogical elements can help achieve a learning objective?	LearningObjective, PedagogicalElement, PedagogicalTechnique
Evaluation	What metrics are used by an evaluation technique?	EvaluationTechnique, EvaluationMetric, EvaluationResult
Feedback & response	What responses originated from a human?	Response, Source, Agent
Learner context	What preferences or challenges does a learner have?	Learner, LearnerPreference, LearningChallenge
Awareness	What is the learner’s current awareness level for a topic or skill?	AwarenessObservation, AwarenessLevel, Topic, Skill

We used the following running use case to scope the ontology. **i.** *Learner may pursue goals that target skills* **ii.** *Learner may have preferences, challenges or awareness over one or more topics or skills* **iii.** *Learning objectives are aligned in relation to learner situations and may be supported by pedagogical techniques or learning resource units* **iv.** *Assessment exercises use evaluation techniques and metrics to produce results that support feedback (with source and temporal information), awareness observations or learning strategies used.* These led to six modeling requirements where APLO needed to represent pedagogical organization, including frameworks, elements and their order; objective alignment across learning objectives, pedagogical techniques and learning resources; evaluation, including assessment exercises, techniques, metrics and results; feedback responses, including source and temporal information; learner context, including goals, preferences, challenges and situations; and learner awareness as topic or skill-scoped information that changes over time.

We used CQs as functional requirements for the ontology by grouping them into families that correspond to the modeling requirements above (see Table 1). These CQ groups guided key notion identification. For instance, CQs about framework membership and ordering motivated PedagogicalFramework, PedagogicalElement, FirstPedagogicalElement, LastPedagogicalElement and next/previous relations. The resulting key notions were then matched to ODPs.

Our ontology engineering method also uniquely shaped the contribution. Modular Ontology Modeling (MOMo) provides a methodology for developing modular ontologies through repeated reviews of use-case definition, CQs, key-notion identification, pattern selection and instantiation, etc., in defined process steps (which can be found in [39]). We used MOMo to structure the development process because the orchestration use case crossed several modeling concerns that needed to remain independently understandable while sharing explicit interfaces. This followed the same pattern-driven approach used in other modular orchestration ontologies, where key notions were matched to reusable patterns when suitable patterns existed and modeled through justified adaptations otherwise. We also largely utilized Modular Ontology Design Library (MODL) – a reusable ODP library that supports modular ontology development – to select reusable patterns for recurring modeling needs [39, 40, 41]. The selected patterns did not function as imported domain ontologies or as classes formally equivalent to APLO classes. Instead, they served as modeling templates whose structures and commitments were specialized to retain domain-specific (education) meanings. We further enforce systematic constraints on each triple, adapting the use of the 17 common axiom templates used in systematic axiomatization¹ [42]. This iterative process resulted in four APLO modules: **a)** pedagogical technique and framework pattern **b)** learning objective pattern **c)** learner evaluation pattern and **d)** learner model pattern. We treated related notions from other ontologies as either alignment candidates or direct reuse, unless an explicit OWL equivalence was asserted. For example, Module, Media and Topic connect to curriculum-facing notions in the CurrKG ontology [7]; or LearnerGoal, Skill and other learner-profile notions relate to

¹The complete axiom selections and their rationales are available in the repository.

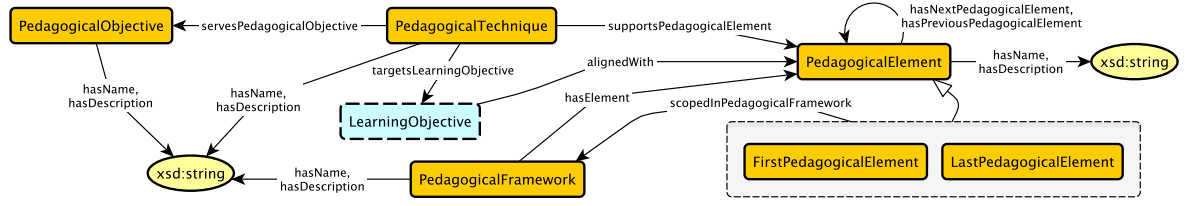


Figure 1: Schema diagram for Pedagogical Technique pattern. The *yellow boxes* indicate the classes, *purple ones* represent the controlled vocabulary, blue ones represent external entities and *yellow ellipses* indicate the datatypes tied to those classes. *Open arrows* indicate SUBCLASSOF relations.

EduCOR-style personalized recommendation concepts [31] We avoided strong owl:equivalentClass commitments when the intended extensions differed and treated them as conceptual alignments for future mapping. All additional information and deliverables for this ontology (i.e., full CQ set, source data, mappings, OWL artifacts and documentation, etc.) are publicly available in our GitHub repository [43].

3.1. Pedagogical Technique and Framework Pattern

Pedagogical technique and framework pattern modeled instructional technique organization, tackling notions such as PedagogicalFramework, PedagogicalElement, PedagogicalTechnique, and PedagogicalObjective alongside their hasElement-isElementOf, targetsLearningObjective or servesPedagogicalObjective relationships (see Figure 1). We primarily adapted PART-WHOLE and SEQUENCE patterns to model PedagogicalFramework as a whole composed of PedagogicalElement instances to optionally have structured order. We kept these commitments separate because framework membership and instructional order model different CQs-Key notions pairs and a framework may contain an element, while next/previous relations specify the ordering within and also out of that scope.

We also distinguished PedagogicalObjective from LearningObjective where a pedagogical objective represented the instructional effect intended by a technique, such as recall, engagement, misconception diagnosis, knowledge transfer or deeper conceptual understanding and a learning objective represented the learner outcome being targeted. This distinction avoided collapsing teacher-side instructional purpose into learner-side outcome, while still allowing pedagogical techniques to target learning objectives, aligning with instructional design works that treat instructional methods, goals and conditions as formal modeling objects [21, 22]. Formally, we defined our core notions in Description Logic (DL) axioms as:

$$\begin{aligned}
 & \exists \text{ hasElement} . \top \sqsubseteq \text{PedagogicalFramework} \\
 & \top \sqsubseteq \forall \text{ hasElement} . \text{PedagogicalElement} \\
 & \text{PedagogicalFramework} \sqsubseteq \exists \text{ hasElement} . \text{PedagogicalElement} \sqcap =1 \text{ hasName} . \text{xsd:string} \\
 & \{\text{First/Last}\} \text{PedagogicalElement} \sqsubseteq \text{PedagogicalElement} \\
 & \text{PedagogicalTechnique} \sqsubseteq \exists \text{ servesPedagogicalObjective} . \text{PedagogicalObjective} \\
 & \exists \text{ targetsLearningObjective} . \top \sqsubseteq \text{PedagogicalTechnique} \\
 & \top \sqsubseteq \forall \text{ targetsLearningObjective} . \text{LearningObjective}
 \end{aligned}$$

Together, these axioms constrain hasElement to relate frameworks to elements; require each framework to contain at least one element and exactly one name; and specialize the first and last sequence positions as elements. They also require each technique to serve a pedagogical objective and constrain targetsLearningObjective to relate techniques to learning objectives.

3.2. Learning Objective Pattern

Learning objective pattern primarily represented the connection between instructional planning and learner context, modeling notions like LearningStrategy and LearningObjective as bridges between LearningSituation, PedagogicalTechnique and Module or Media entities (see Figure 2).

Specifically, LearningStrategy was modeled to represent a named instructional configuration that used pedagogical techniques to pursue learning objectives and to capture the specific instructional strategy component that is scoped to the learner context, bound together in a tailored teaching plan. As such, it heavily adapted DESCRIPTION-SITUATION and NAME-STUB patterns for capturing scoped details of this bridging concept while also connecting to curriculum-facing entities.

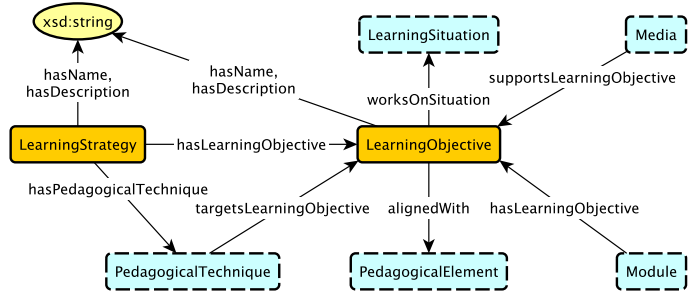


Figure 2: Schema diagram for Learning Objective pattern.

LearningSituation represented the learner’s contextual state relevant to an objective, such as prior topic knowledge or situational constraints. In our design, LearningStrategy is connected to LearningObjective through hasLearningObjective and then to LearningSituation through worksOnSituation relations. This connected APLO to prior instructional-design and objective-modeling work on instructional-design theories with ontologies that represented methods, conditions and learning designs as formal knowledge; an ontology-based modeling framework for educational-technology design; and learning objective ontological alignment across educational and interactive contexts [21, 22, 25]. We used LearningObjective as an interface across pedagogy, learner context, curriculum resources and evaluation rather than as an isolated objective statement, distinguishing from prior works. We also directly reused some CurrKG classes, preserved identity across the two ontologies, avoided the need to maintain parallel module or resource hierarchies and gave the entities instructional roles within APLO. Henceforth, we formally defined some key notions in DL axioms as:

$$\begin{aligned}
 &\text{LearningStrategy} \sqsubseteq =1 \text{ hasName.xsd:string} \sqcap \exists \text{ hasLearningObjective.LearningObjective} \\
 &\quad \sqcap \exists \text{ hasPedagogicalTechnique.PedagogicalTechnique} \\
 &\text{LearningObjective} \sqsubseteq =1 \text{ hasName.xsd:string} \sqcap \exists \text{ developsSkill.Skill} \\
 &\quad \text{Module} \sqsubseteq \exists \text{ hasLearningObjective.LearningObjective} \\
 &\quad \text{Media} \sqsubseteq \exists \text{ supportsLearningObjective.LearningObjective} \\
 &\exists \text{ worksOnSituation.T} \sqsubseteq \text{LearningObjective} \\
 &\quad \text{T} \sqsubseteq \forall \text{ worksOnSituation.LearningSituation} \\
 &\exists \text{ alignedWith.T} \sqsubseteq \text{LearningObjective} \\
 &\quad \text{T} \sqsubseteq \forall \text{ alignedWith.PedagogicalElement} \\
 &\quad \text{T} \sqsubseteq \forall \text{ hasLearningObjective.LearningObjective}
 \end{aligned}$$

Together, these axioms require each strategy to have exactly one name, at least one learning objective and at least one technique. Each learning objective has exactly one name and develops at least one skill. Modules and media participate through explicit objective relations. The remaining axioms constrain worksOnSituation, alignedWith and hasLearningObjective to their intended domains or ranges without requiring every objective to be immediately contextualized or aligned.

3.3. Learner Evaluation Pattern

Learner evaluation pattern represented assessment activity, evaluation techniques, measured results, feedback and response provenance, including EvaluationAssessmentExercise, EvaluationTechnique, EvaluationTechniqueType, EvaluationMetric, EvaluationResult, LearningFeedback, LearningFeedbackType, Response, Source and WorkloadLevel notions (see Figure 3). This pattern required the richest ODP composition. We decomposed and distinguished multiple aspects of Response notion by adapting OBSERVATION pattern to model LearningFeedback as evidence about learner performance or understanding and QUANTITY pattern to separate EvaluationMetric from EvaluationResult, so that a metric represented the kind or unit of measurement while a result represented a measured value. Additionally, we minimally adapted from REPORTING-EVENT and PROVENANCE to model broader Response as a delivered output, rather than reifying a separate

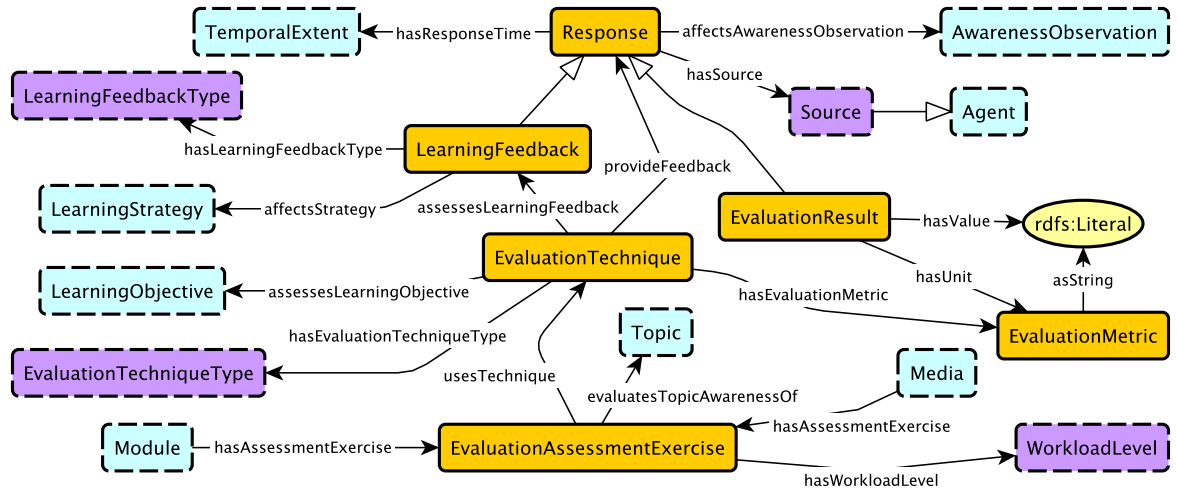


Figure 3: Schema diagram for Learner Evaluation pattern.

communication event, and source and temporal information with TEMPORALEXTENT and AGENT-ROLE patterns to represent sources such as teachers, tutoring agents or learners. These choices kept the pattern lightweight while allowing it to inherit common provenance semantics.

As no single ODP captured the complete pattern requirement, OBSERVATION captured learner evidence, QUANTITY the metric/result semantics, REPORTING-EVENT the delivered response, PROVENANCE and AGENT-ROLE the source of that response broadly to model a complex interaction. This allowed APLO to answer narrow CQs about which technique assessed an objective, which metric was used, what feedback resulted, when a response was delivered and whether the response originated from a human or another source. This pattern is also aligned with OLM concerns where evaluation results and feedback can provide evidence for learner-state interpretation, while response provenance helps explain where feedback came from. As such, we treated related assessment and learner-evidence notions in prior work as conceptual alignments rather than formal equivalences, because APLO used them specifically for orchestration among evaluation, feedback and awareness. Some main key notions in DL axioms are formally defined as follows:

$$\begin{aligned}
 &\{ \text{LearningFeedback} / \text{EvaluationResult} \} \sqsubseteq \text{Response} \\
 &\text{Source} \sqsubseteq \text{Agent} \\
 &\text{EvaluationAssessmentExercise} \sqsubseteq \exists \text{ usesTechnique.EvaluationTechnique} \\
 &\quad \sqcap \exists \text{ evaluatesTopicAwarenessOf.Topic} \sqcap =1 \text{ hasWorkloadLevel.WorkloadLevel} \\
 &\text{EvaluationTechnique} \sqsubseteq \exists \text{ hasEvaluationMetric.EvaluationMetric} \\
 &\quad \sqcap \exists \text{ hasEvaluationTechniqueType.EvaluationTechniqueType} \\
 &\text{EvaluationResult} \sqsubseteq \exists \text{ hasUnit.EvaluationMetric} \sqcap \exists \text{ hasValue.rdfs:Literal} \\
 &\text{LearningFeedback} \sqsubseteq =1 \text{ hasLearningFeedbackType.LearningFeedbackType} \\
 &\text{Response} \sqsubseteq =1 \text{ hasSource.Source} \sqcap \leq 1 \text{ hasResponseTime.TemporalExtent} \\
 &\exists \text{ provideFeedback.T} \sqsubseteq \text{EvaluationTechnique} \\
 &\quad \text{T} \sqsubseteq \forall \text{ provideFeedback.Response} \\
 &\exists \text{ affectsAwarenessObservation.T} \sqsubseteq \text{Response} \\
 &\quad \text{T} \sqsubseteq \forall \text{ affectsAwarenessObservation.AwarenessObservation}
 \end{aligned}$$

Together, these model: evaluation results and learning feedback as provenance-bearing responses and constrain response sources to agents; Assessment exercises to use an evaluation technique, evaluate awareness of at least one topic and have exactly one workload level; Evaluation techniques to use a metric and have a controlled type, while evaluation results carry a value and metric; A feedback item to have exactly one feedback type; and each response to have exactly one source and at most one response time. The remaining axioms constrain optional links from evaluation techniques to responses and then awareness.

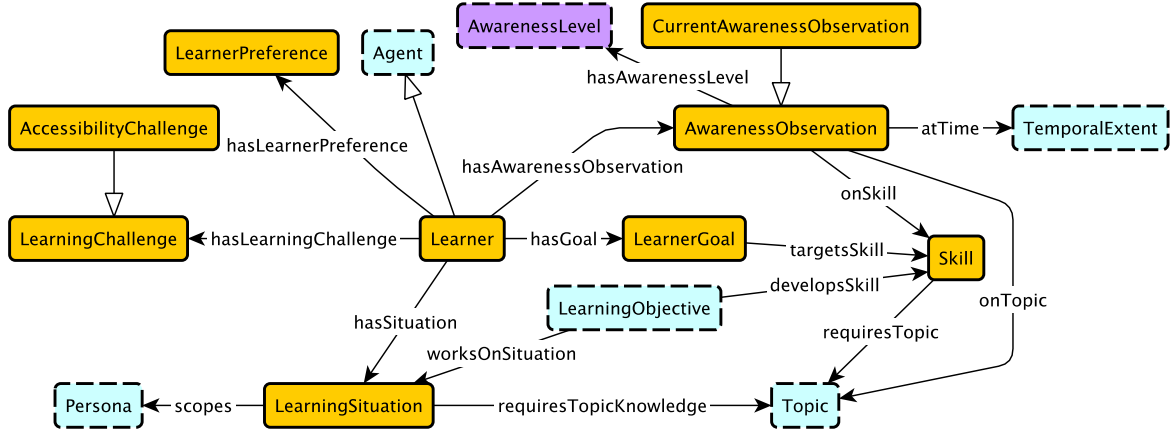


Figure 4: Schema diagram for Learner Model pattern.

3.4. Learner Model Pattern

Learner model pattern, largely motivated from various state-of-the-art OLM work, has modeled learner-specific information needed for adaptive orchestration with main notions such as *Learner*, *LearnerGoal*, *LearnerPreference*, *LearningChallenge*, *AwarenessObservation*, *AwarenessLevel*, and then further scopes it with *LearningSituation*, *Skill* and *Topic*. *Learner* remains a central entity with: *LearnerGoal* targeting *Skill*; *LearningSituation* considering prior *Topic* knowledge; and *AwarenessObservation* scoped to *Topic* or *Skill* with *AwarenessLevel* and temporal extents. One main design decision was to model awareness as a temporally scoped SOSA-like *Observation* rather than as a static learner attribute. This followed the OLM view that learner-state representations are useful when they can be inspected, interpreted and connected to evidence [33, 34, 35, 36]. Henceforth, in APLO, an awareness—scoped to topic, skill, level and time—supported the orchestration use case because later pedagogical decisions may depend on current awareness, low-awareness topics or relations between learner goals and skills. External equivalence and alignments included *Learner*, modeled as an *AGENT*, to learner-profile or user-profile notions; *LearnerGoal* to learning-goal notions; and *Skill* to skill notions in personalized educational and career-oriented ontologies such as EduCOR [31]. However, we broadened these concepts for adaptive orchestration rather than recommendation alone; therefore, we represented them as APLO classes and treated external correspondences as future mapping candidates. Formally, we defined some main semantic commitments in DL axioms as:

$$\begin{aligned}
& \text{Learner} \sqsubseteq \text{Agent} \sqcap \exists \text{hasSituation}.\text{LearningSituation} \\
& \exists \text{hasGoal}.\top \sqsubseteq \text{Learner} \\
& \top \sqsubseteq \forall \text{hasGoal}.\text{LearnerGoal} \\
& \exists \text{hasLearnerPreference}.\top \sqsubseteq \text{Learner} \\
& \top \sqsubseteq \forall \text{hasLearnerPreference}.\text{LearnerPreference} \\
& \exists \text{hasLearningChallenge}.\top \sqsubseteq \text{Learner} \\
& \top \sqsubseteq \forall \text{hasLearningChallenge}.\text{LearningChallenge} \\
& \text{LearnerGoal} \sqsubseteq \forall \text{targetsSkill}.\text{Skill} \\
& \text{AccessibilityChallenge} \sqsubseteq \text{LearningChallenge} \\
& \text{AwarenessObservation} \sqsubseteq =1 \text{hasAwarenessLevel}.\text{AwarenessLevel} \sqcap \leq 1 \text{atTime}.\text{TemporalExtent} \\
& \text{AwarenessObservation} \sqsubseteq \exists \text{onTopic}.\text{Topic} \sqcup \exists \text{onSkill}.\text{Skill} \\
& \text{CurrentAwarenessObservation} \sqsubseteq \text{AwarenessObservation} \\
& \text{Skill} \sqsubseteq \exists \text{requiresTopic}.\text{Topic}
\end{aligned}$$

Together, these axioms model Learners as *AGENTS* participating in *LearningSituations* while allowing goals, preferences and challenges to be included. Each *LearnerGoal* may target a *Skill*. Every *AwarenessObservation* has exactly one level, at most one *TEMPORALEXTENT* and concerns at least one *Topic* or *Skill*. *CurrentAwarenessObservation* specializes *AwarenessObservation*, while each *Skill* requires at least one *Topic*.

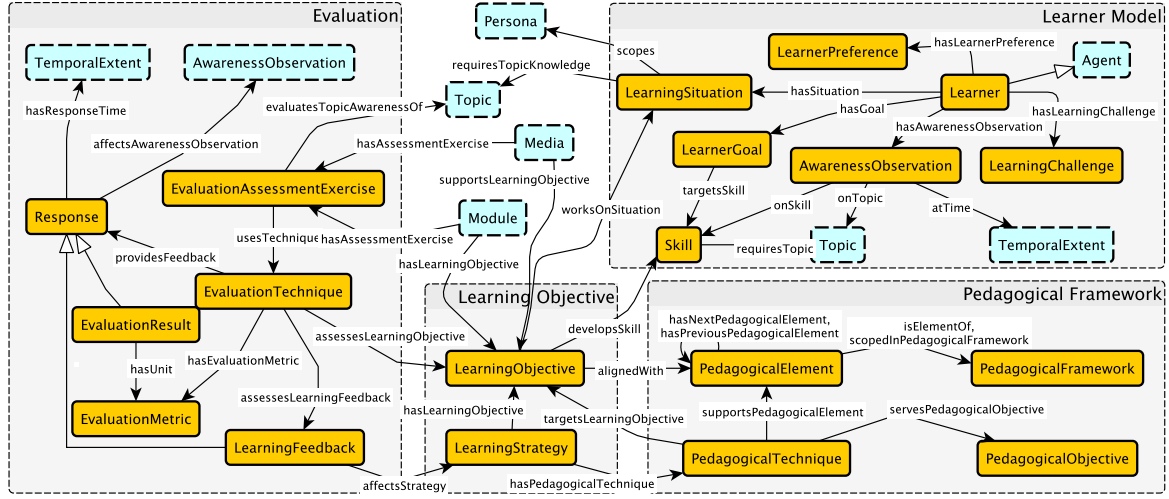


Figure 5: Abridged schema diagram of all patterns concatenated.

After instantiating all the patterns, we assembled them through explicit interface classes rather than through a single orchestration class to obtain an abridged view of the entire APLO schema (see Figure 5). The assembled ontology therefore treated orchestration as a cross-module representational capability where each pattern (doubling as a module) remained independently understandable but their adaptive use depended on shared interface classes and properties. This also allowed APLO to reuse curriculum-facing entities without duplicating a full curriculum ontology. All ontology artifacts, schema diagrams, documentation (CQs, key notions, alignments, equivalences and supplementary materials) are available to extend, adapt and reproduce in our repository [43].

4. Materialization & Evaluation

APLO was evaluated as both an ontology artifact and schema for materialized RDF data to check whether its modules, axioms, alignments and materialized instances supported the use case and information dependencies. As such, the ontology was subjected to different checks:

Structural Check. This was conducted iteratively with OOPS! [44] for each pattern, and over the whole ontology, identifying common ontology pitfalls and pertinent structural problems (e.g., missing axioms or property characteristics) until all issues were resolved. This acted as a structural quality gate rather than as a complete semantic validation to ensure the final ontology artifact complies with best practices at OOPS! level and is relevant to the intended modeling scope. The most common revisions concerned missing annotations, domain & range constraints and inverse relations.

Materialization. Since most of the orchestration data was system process-based and due to the unawareness of the availability of such complete public datasets that jointly covered pedagogical frameworks, objectives or learners' context, evaluation results, feedback or awareness by the authors, we proceeded with constructing a mock synthetic dataset pertaining to the domain of KG education (adapted from OpenKGC curriculum [45]). All synthetic files are constructed according to individual use cases across APLO modules, in a normalized fashion, resulting in 22 files. This is then used alongside Foundry [46], an open-source RDF KG materialization tool, to create RDF graphs from normalized data sources and mapping scripts for each pattern, resulting in 6836 triples. All synthetic data and mapping scripts used for materialization are also available in the repository [43]. This step was taken to check APLO at the representational coverage level and whether it could serve as a schema for integrated orchestration data across pedagogy, objectives, learner information and curriculum resources.

Use Case Evaluation. Using these materialized instances and our ontology OWL structures, we loaded them all into a triplestore (i.e., Apache Jena Fuseki [47]). To check if the ontology sufficiently captures all the information and semantics it is designed to do, we formulated SPARQL queries for each of the 94 CQs (example CQ-query pair shown in Fig 6) and executed them over the triplestore to check for execution and results retrieved. A CQ was counted as supported when its corresponding query executed successfully and returned results consistent with the expected use-case data. This process of executing and verifying each CQ intent ensured that the APLO ontology sufficiently captured the semantics it is designed to do as it attained 100% CQ coverage [7, 39, 48, 49, 50]. It also exposed whether materialized data exercised the intended module interfaces. For instance, queries about assessment exercises minimally required consistent links among *EvaluationAssessmentExercise*, *EvaluationTechnique*, *EvaluationMetric*, *LearningObjective* and *Topic*; or the ones about awareness required *Learner*, *AwarenessObservation*, *AwarenessLevel*, *Topic* and *Skill*. Thus, this validation tested not only isolated classes but also cross-module interfaces.

Together, these validations ensured the ontology satisfied its artifact-level requirements, supporting APLO as a reusable symbolic schema for adaptive pedagogical orchestration representational gap. However, the CQs require downstream system integration and educational evaluation to establish that APLO improves learning outcomes or selects optimal pedagogical actions for a complete NaPA, which we leave for future work.

5. Pattern Composition and Design Trade-offs

APLO design case showed that adaptive pedagogical orchestration could not be modeled by applying a single ODP as modeling problem crossed instructional structure, learning and learner nuances, making pattern composition the main design issue. Though some parts mapped clearly to established ODPs (e.g., pedagogical framework to *PART-WHOLE & SEQUENCE* for ordered membership), others required tighter composition. Learner evaluation pattern provides the clearest example where *OBSERVATION* captured evidence, *QUANTITY* the separation between evaluation metrics and measured results, *REPORTING-EVENT* the delivered response, *PROVENANCE* the origin and traceability, while *AGENT-ROLE* the source type. This complexity was necessary because assessment, feedback and response are closely related but not identical and collapsing them into one generic feedback would simplify but also remove the ability to infer the technique used, metric applied, result measured, response origin and its delivery.

A second design trade-off concerned modularity where APLO treated orchestration as a cross-module representational capability which kept the patterns independently understandable, while allowing adaptive use through shared interface classes such as *LearningObjective*, *LearningSituation*, *Topic* or *Media* etc. Though the benefit was reuse (e.g., one can reuse the learner model pattern without adopting the full evaluation pattern), the cost was that several important meanings and semantics live at pattern boundaries. Similarly, a third concerned external reuse and alignment. APLO reused or connected to curriculum-facing notions such as *Module*, *Media* or *Topic* where *CurrKG* already provided the relevant curriculum-material structures, or notions such as *LearnerGoal*, *Skill* and learner-profile concepts as related to *EduCOR*-style personalized models [7, 31]. However, the choice of avoiding broad *owl:equivalentClass* commitments mattered because APLO used several notions for orchestration and not for recommendation or learner-profile representation, treating many correspondences as conceptual alignments or future mapping candidates rather than as traditional equivalence axioms. This conservative mapping reduced the risk of importing unintended commitments from external ontologies.

CQ1: Which instructional steps can support a learner working toward a given learning objective?

```
SELECT DISTINCT ?learner ?goal ?learningObjective ?learningObjectiveLabel ?
instructionalStep ?instructionalStepLabel
WHERE {
  ?learner a aplo-ont:Learner ;
    aplo-ont:hasGoal ?goal .
  ?goal aplo-ont:targetsSkill ?skill .
  ?learningObjective a aplo-ont:LearningObjective ;
    aplo-ont:developsSkill ?skill ;
    aplo-ont:alignedWith ?instructionalStep .
  ?instructionalStep a aplo-ont:PedagogicalElement .
  OPTIONAL { ?learningObjective aplo-ont:hasName ?learningObjectiveLabel . }
  OPTIONAL { ?instructionalStep aplo-ont:hasName ?instructionalStepLabel . }
}
ORDER BY ?learner ?learningObjective ?instructionalStep
```

Figure 6: Example CQ-to-SPARQL validation query.

APLO’s learner model exposed another modeling choice in representing awareness as a temporally scoped observation rather than as a static learner attribute. Though this decision followed OLMs’ view for inspectable, interpretable and evidence-connected modeling, the cost was that it then needed explicit support for current observations and temporal extents, making it more expressive but also necessitating more complete data for downstream use [33, 34, 35, 36]. Furthermore, evaluation (see Sec 4) supported APLO at the artifact level with mitigating structural checks, realizing with materialization and design and use-case compliance with CQ-SPARQL validation. Although these results support the representational fitness for adaptive pedagogical orchestration, they do not show that APLO improves the intended modeling motivation claims for adaptive educational systems but stand as ontology modeling process validators, requiring empirical educational validation for outcome-level evaluation.

The main limitation of the current work is therefore the use of synthetic orchestration data. It exercised the schema and supported CQ validation but does not consider the messiness of classroom practice, learner behavior or longitudinal adaptation. A second one is that external alignments remain mostly conceptual. Future versions should add explicit mapping axioms where the intended extensions match, and weaker mappings such as `skos:closeMatch` or `rdfs:seeAlso` when partial overlap exists. A third limitation is that APLO currently represents the information dependencies needed for orchestration, but not the decision policy that chooses the next pedagogical action. Future work should connect APLO to real educational data, expert review, runtime tutoring workflows and NaPAs.

6. Conclusion

We presented APLO, a modular ontology for representing adaptive pedagogical learning orchestration, addressing the modeling gap between curriculum-centered educational knowledge representations and the process-aware representations needed for adaptive instruction. It modeled orchestration as a cross-module representational capability rather than as a single monolithic class, integrating four interoperable modules for pedagogical technique modeling, learning objectives, learner evaluation and learner context. The ontology was evaluated at the artifact level through structural quality checking, RDF materialization and SPARQL-based CQ validation and the result therefore provides a reusable symbolic schema for connecting curriculum-facing entities, pedagogical structures, learner state, evaluation evidence and feedback processes, while clarifying external reuse and alignment role.

Future Work. Apart from the directions discussed in Section 5, future work should extend these mappings with explicit or weaker SKOS-style alignments where appropriate, evaluate and reiterate design with real educational data and ontology expert & educator review and assess downstream quality and adaptability in intelligent tutoring workflows.

Acknowledgments

Chris Davis Jaldi and Cogan Shimizu acknowledge support from the National Science Foundation under awards #2537485 “Prototyping a new Knowledge Resource for modern AI (Proto-KAI)”, #2333532 “Proto-OKN Theme 3: An Education Gateway for the Proto-OKN (EduGate)”, and #2514820 “EAGER: Feasibility study for a Translational Institute on Knowledge Axiomatization (TIKA)” and from the US Department of Education under award #P116J250669 “Strengthening American Competitiveness with AI Education (SACAIEda).”

Declaration on Generative AI

The author(s) have not employed use of Generative AI in preparation of manuscript.

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